UF3 KOKKOS Tensor Core Implementation

Mitch Murphy

April 9, 2026

Matrix and Tensor Operations for the UF3 Potential

The provided C++ code evaluates an empirical potential (likely a machine-learning model like UF3) utilizing cubic B-splines for 2-body and 3-body interactions.

1. Distance Polynomial Vectors

First, we define the polynomial distance vectors used throughout the code to evaluate the splines. For any pair distance r, we define the 4×1 position vector $\mathbf{p}(r)$ and the 3×1 derivative vector $\mathbf{p}'(r)$:

$$\mathbf{p}(r) = \begin{bmatrix} 1 \\ r \\ r^2 \\ r^3 \end{bmatrix}, \quad \mathbf{p}'(r) = \begin{bmatrix} 1 \\ r \\ r^2 \end{bmatrix} \tag{1}$$

2. Two-Body Interactions

The 2-body scalar force and energy are computed using a set of spline coefficients. Let C_{2b} be the 4×4 coefficient matrix (`constants_2b`) and C'_{2b} be the 3×3 derivative coefficient matrix (`constants_2b_deri`).

2-Body Energy:

$$E_{2b} = \mathbf{1}_4^T C_{2b} \mathbf{p}(r_{ij}) \tag{2}$$

(where $\mathbf{1}_4$ is a 4×1 vector of ones).

2-Body Scalar Force:

$$F_{2b} = \mathbf{1}_3^T C'_{2b} \mathbf{p}'(r_{ij}) \tag{3}$$

2-Body Vector Force: Using the displacement vector $\Delta\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j$, the force applied to atom i is:

$$\mathbf{f}_i^{(2b)} = -\frac{F_{2b}}{r_{ij}}\Delta\mathbf{r}_{ij} \tag{4}$$

3. Three-Body Interactions: Basis Vectors

For a triplet of atoms (i, j, k), the code computes spline basis vectors for each pair distance (r_{ij}, r_{ik}, r_{jk}). Let C_{ij} be the 4×4 coefficient matrix (`cc_3b_ij`) and C'_{ij} be the 3×3 derivative matrix (`cc_3b_deri_ij`).

The 4×1 basis vectors ($\mathbf{b}$) and 3×1 derivative basis vectors ($\mathbf{b}'$) are obtained via matrix-vector multiplication:

$$\mathbf{b}_{ij} = C_{ij}\mathbf{p}(r_{ij}), \qquad \mathbf{b}'_{ij} = C'_{ij}\mathbf{p}'(r_{ij}) \tag{5}$$

$$\mathbf{b}_{ik} = C_{ik}\mathbf{p}(r_{ik}), \qquad \mathbf{b}'_{ik} = C'_{ik}\mathbf{p}'(r_{ik}) \tag{6}$$

$$\mathbf{b}_{jk} = C_{jk}\mathbf{p}(r_{jk}), \qquad \mathbf{b}'_{jk} = C'_{jk}\mathbf{p}'(r_{jk}) \tag{7}$$

4. Three-Body Interactions: Tensor Contractions

The nested loops representing `triangle_eval` variables are tensor contractions. The coefficients are stored in 3-dimensional tensors, which we can represent as sums of bilinear forms evaluated over matrix slices W_l.

Three-Body Energy (`triangle_eval0`):
Let $W^{(E)}$ be the $4 \times 4 \times 4$ energy coefficient tensor (`n3b_coeff_array`). The energy is the contraction of the basis vectors along the three axes:

$$E_{3b} = \sum_{l=0}^{3} b_{ij,l}\left(\mathbf{b}_{ik}^T W_l^{(E)} \mathbf{b}_{jk}\right) \tag{8}$$

Three-Body Scalar Forces (`triangle_eval1, 2, 3`):
The code evaluates the derivative components by contracting the derivative basis vector of one pair with the standard basis vectors of the other two pairs.

For the ij pair (`triangle_eval1`), using the 4×4 matrix slices $W_l^{(ij)}$ from `coeff_for_der_ij`:

$$T_1 = \sum_{l=0}^{2} b'_{ij,l}\left(\mathbf{b}_{ik}^T W_l^{(ij)} \mathbf{b}_{jk}\right) \tag{9}$$

For the ik pair (`triangle_eval2`), using the 3×4 matrix slices $W_l^{(ik)}$ from `coeff_for_der_ik`:

$$T_2 = \sum_{l=0}^{3} b_{ij,l}\left((\mathbf{b}'_{ik})^T W_l^{(ik)} \mathbf{b}_{jk}\right) \tag{10}$$

For the jk pair (`triangle_eval3`), using the 4×3 matrix slices $W_l^{(jk)}$ from `coeff_for_der_jk`:

$$T_3 = \sum_{l=0}^{3} b_{ij,l} \left(\mathbf{b}_{ik}^T W_l^{(jk)} \mathbf{b}'_{jk} \right) \tag{11}$$

5. Three-Body Vector Forces

Finally, the scalar quantities T_1, T_2, and T_3 are projected back onto the cartesian unit vectors between the atoms. Let the unit displacement vectors be defined as $\hat{\mathbf{r}}_{ji} = \frac{\mathbf{r}_j - \mathbf{r}_i}{r_{ij}}$, $\hat{\mathbf{r}}_{ki} = \frac{\mathbf{r}_k - \mathbf{r}_i}{r_{ik}}$, and $\hat{\mathbf{r}}_{kj} = \frac{\mathbf{r}_k - \mathbf{r}_j}{r_{jk}}$.

The intermediate force vectors are:

$$\mathbf{f}_{ij} = T_1 \hat{\mathbf{r}}_{ji} \tag{12}$$

$$\mathbf{f}_{ik} = T_2 \hat{\mathbf{r}}_{ki} \tag{13}$$

$$\mathbf{f}_{jk} = T_3 \hat{\mathbf{r}}_{kj} \tag{14}$$

The total force distributed to atoms i, j, and k from this triplet interaction is simply the linear combination of these intermediate vectors:

$$\mathbf{F}_i = \mathbf{f}_{ij} + \mathbf{f}_{ik} \tag{15}$$

$$\mathbf{F}_j = -\mathbf{f}_{ij} + \mathbf{f}_{jk} \tag{16}$$

$$\mathbf{F}_k = -\mathbf{f}_{ik} - \mathbf{f}_{jk} \tag{17}$$

$$
\begin{bmatrix}
1 & 2 & -1 & 3 & 4 \\
0 & 1 & 3 & -2 & -1 \\
0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0
\end{bmatrix}
$$